Project Report: Enhancing Windows Security with Logon Monitoring and Reporting

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1. Project Overview & Objective

- **Task**: Enhance the security posture of a Windows-based enterprise system to detect and mitigate unauthorized login attempts and malicious command executions.
- **Action**: Implemented advanced audit configurations using Group Policy, simulated a brute-force attack to generate security logs, and utilized PowerShell to automate the extraction of threat data into an HTML report.
- **Result**: Established a proactive monitoring environment that provides real-time visibility into system access and automated reporting for rapid incident response.

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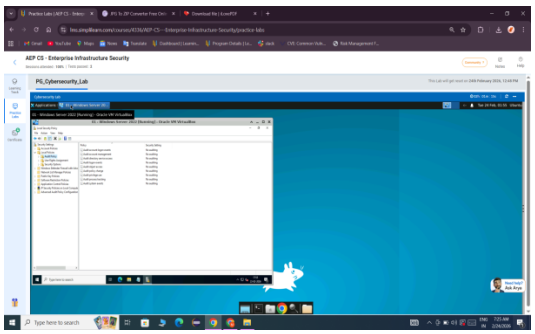
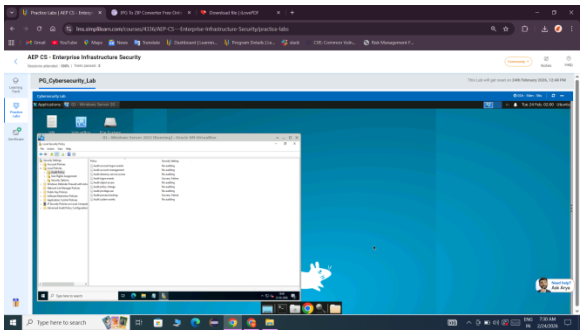
2. Advanced Audit Configuration (Group Policy)

- **Task**: Enable detailed tracking of user access and executed processes to ensure all potential security events are recorded.
- **Action**: Accessed the Local Security Policy manager (`secpol.msc`) and navigated to the **Audit Policy** settings.
- **Configuration**:
  - Enabled **Audit logon events** for both *Success* and *Failure* to track password guessing attempts.
  - Enabled **Audit process tracking** for both *Success* and *Failure* to document executed commands.
- **Result**: Successfully transitioned the server from a default "No auditing" state to an active monitoring state, turning on the system's "security cameras."
- **Evidence Files**:
-  

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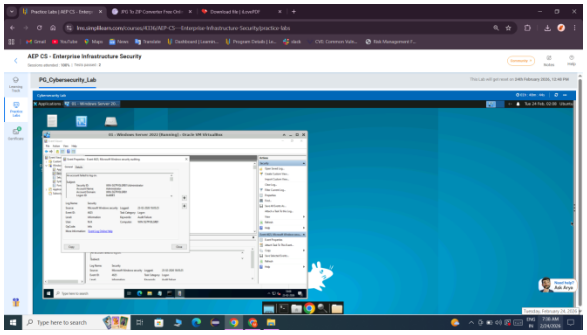
3. Threat Simulation & Log Analysis (Event Viewer)

- **Task**: Validate that the new audit policies actively capture unauthorized access attempts.
- **Action**: Executed a simulated attack using the Command Prompt (`runas /user:Administrator cmd`) by intentionally inputting incorrect administrator credentials.
- **Verification**: Opened **Event Viewer** (`eventvwr.msc`) and navigated to the Windows Security logs to hunt for the generated alerts.
- **Result**: Successfully located **Event ID 4625 (Audit Failure)**, confirming the system is actively logging and detailing failed authentication attempts.
- **Evidence File**:
- 

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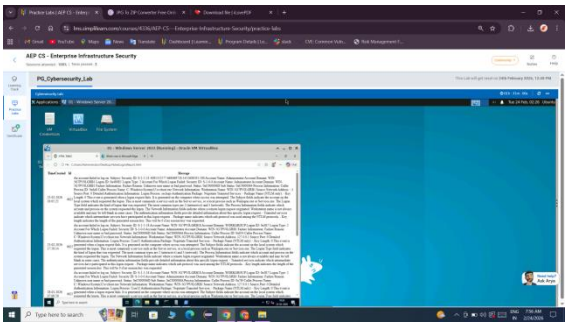
4. Automated Security Reporting (PowerShell & HTML)

- **Task**: Streamline the incident response process by automating the analysis and reporting of failed logon events.
- **Action**: Executed an advanced PowerShell script with administrative privileges to query the Security logs specifically for Event ID 4625.
- **Command Executed**: `Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 5 | Select-Object TimeCreated, Id, Message | ConvertTo-Html | Out-File "$env:USERPROFILE\Desktop\FailedLoginsReport.html"`
- **Result**: The script successfully parsed the raw event data and automatically generated a clean, readable HTML web report directly to the desktop.
- **Evidence File**:
- 

Conclusion

The implementation was highly successful. By leveraging native Windows Group Policy Management, the system's logging capabilities were significantly enhanced to capture critical security events. Furthermore, utilizing PowerShell to automate the generation of HTML reports from Event Viewer data ensures that security analysts can rapidly identify and respond to brute-force threats, strengthening the overall enterprise security posture.